

Fig. 3: Working of a typical model rocket; a typical model rocket (inset)



Fig. 4: Model rocket igniter



Fig. 5: Model rocket launch controller

The tip of the igniter consists of a very thin wire coated with a suitable chemical compound that heats up, for example, pyrogen. The thin wire (also known as bridge wire) heats up when sufficient electricity flows through it, which causes the pyrogen to combust. This initiates the burning of the solid rocket engine propellant.

Note that, model rocket ignition calls for as little as 50mA to as much as 50A, depending on the igniter.

Model rocket launch controller

A launch controller is designed to do one thing—ignite the rocket engine, that is, to supply enough power to the igniter to fire the rocket engine. However, a launch controller should have some safety features like accidental ignition protection, igniter continuity indication, efficient energy-delivery mechanism and so on. Fig. 6 reveals that the launch controller circuitry is very simple.

When the security key is re-

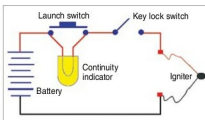


Fig. 6: Basic wiring diagram of a model rocket launch controller



Fig. 7: DIY kit of an advanced model rocket launch controller



Fig. 8: Model rocket motor as a replacement part



Fig. 9: Components of a rocket controller

moved from the key lock switch (arm switch), DC supply from the battery is not available to the igniter and the continuity indicator, so nothing happens. When the security key is in place, DC supply is fed to the continuity indicator through the key lock switch and the igniter.

If the igniter is connected (and in good condition), the continuity indicator illuminates. When the security key is in place, the igniter is good, and the launch switch is pressed, DC supply is extended to the igniter through the launch switch. This allows electric current to pass through the igniter to heat it up and start the rocket. Note that, because of the tricky wiring method (key lock switch in series with launch switch), ignition cannot take place just by pressing the launch switch alone.

Usually, the key lock switch is a metal peg with a tiny rubber ring.

To prevent an accidental launch, the security key must be inserted into the key lock switch and held down while pressing the launch switch. The key automatically pops out when the user releases the applied pressure.

Try it yourself

Today's model rocket launch systems provide safe and reliable means of launching model rockets from a distance. Although single-pad launch systems are very common, you can opt for multi-pad launch systems for flying many model rockets in a rapid sequence.

For do-it-yourself (DIY) hobbyists, it is easy to build their own simple (or feature-packed) model rocket launch controllers with the help of tried-and-tested circuits/DIY kits available elsewhere on the Internet. Readymade model rocket launch controller kits are also available.

Since rocket engine igniters consume many amps of current to ignite, powerful batteries and connection cables are crucial for ensuring total safety and reliability. So always keep an eye on accessories like battery, cables and connectors you select for your model rocket launch system.

About 90 per cent of launch failures can be avoided by following these simple rules:

1. Always work with the recommended battery/power supply
2. Use sufficiently-thick wires/cables for interconnections between the battery, launch controller and igniter
3. Choose high-quality igniter alligator clips (igniter is usually connected to the launch controller using alligator clips)
4. If necessary, clean battery terminals and alligator clips before each flight

Warning! While igniters for model rockets are quite safe, it is still a good idea to reduce any possibility of unexpected voltage potentials and stray currents through igniters. This is especially important during connection of igniters when you are physically close to the rockets. **EFY**